

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed systems, data infrastructure, and AI/ML pipelines at scale. Owned end-to-end systems processing **billions of daily requests** at Google and YouTube. Led teams shipping products to hundreds of millions of users. Partners with product, data science, and SRE on reliability, performance, and customer outcomes. Current focus on **AI agent workflows**, **LLM pipelines**, and **TypeScript** product development. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: JavaScript, HTML, CSS, SQL

Distributed Systems & Infrastructure: Python, C++, Go, Kubernetes, GCP, distributed systems, on-call, kernel level

AI/ML & Data Engineering: Python, data pipelines, LLM workflows, AI agents, data modeling, ML infrastructure

Full-Stack Product Development: TypeScript, Python, React, APIs, database design, end-to-end

Platform & Developer Experience: CI/CD, incident response, technical leadership, code review

Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed and scaled video infrastructure serving billions of users, owning systems from ingestion through playback delivery with **99.99% uptime**.
- Built **distributed data pipelines** processing real-time engagement metrics, enabling content recommendation improvements that increased watch time by double digits.
- Partnered with ML teams to deploy **AI/ML pipelines** for content moderation, reducing manual review workload by 40%.
- Led cross-functional initiatives with product and data science to ship features reaching 500M+ monthly active users.
- Established reliability standards and **on-call** processes for video platform infrastructure handling 10B+ daily requests.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built core platform infrastructure supporting live commerce for creators, processing **\$500M+ in annual GMV** across millions of transactions.
- Architected **data pipelines** and semantic models structuring health technology evaluations for enterprise customers.
- Developed full-stack **TypeScript** and **Python** features for complex ROI decision tooling used by health system analysts.
- Partnered with customers directly to translate ambiguous requirements into shipped product solutions with measurable adoption.
- Owned **end-to-end** project delivery from problem discovery through production deployment and monitoring.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers building search infrastructure serving billions of queries daily with sub-100ms latency requirements.
- Designed **distributed systems** handling **billions of daily requests** across global infrastructure with **99.9% reliability**.

- Led platform migration from monolith to microservices using **Kubernetes** and **GCP**, reducing deployment cycle from monthly to weekly.
- Built **AI/ML pipelines** for ad ranking and user personalization, improving engagement metrics by 15%.
- Established SRE practices across distributed teams, including **on-call** rotations, incident response, and postmortem processes.
- Partnered with product and design to define and ship major product initiatives across core search surfaces.

Software Engineer | Google | 02/2011 - 02/2015

- Developed backend systems for search ranking using **C++** and **Python**, processing millions of queries per second.
- Built **data pipelines** for training set generation feeding ML models serving search results.
- Optimized query processing latency through **C++** performance profiling and **kernel level** system call analysis.
- Participated in production **on-call** rotation responding to incidents affecting search quality and availability.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and shipped features for Windows and Office platforms using **C++** and **C#**.
- Conducted design reviews, code reviews, and architecture discussions for team deliverables.
- Collaborated with program managers and testers on requirements definition and validation.
- Participated in **on-call** rotation supporting production environments for enterprise customers.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal tooling using **C++** and .NET frameworks under mentorship from senior engineers.
- Participated in team meetings, design reviews, and Agile development processes.

Education

B.S. Honors Degree

Virginia Tech | 2004 - 2007